

RLHF

DATA 443/643 · Week 9, note 9.2

Week 9

American University

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Ideas from Stanford CS224N 2025 L10 (InstructGPT: SFT, RM, then KL-leashed RL). Original slides; that course is not copied.

Lecture plan

- 1 Three stages; only the last is RL
- 2 A KL leash to SFT
- 3 A two-word KL cartoon
- 4 Why people look past PPO

Key goal: $\beta=0$ hacks the RM. π_{ref} is the SFT model, not the base LM.

First time: what / why / formula

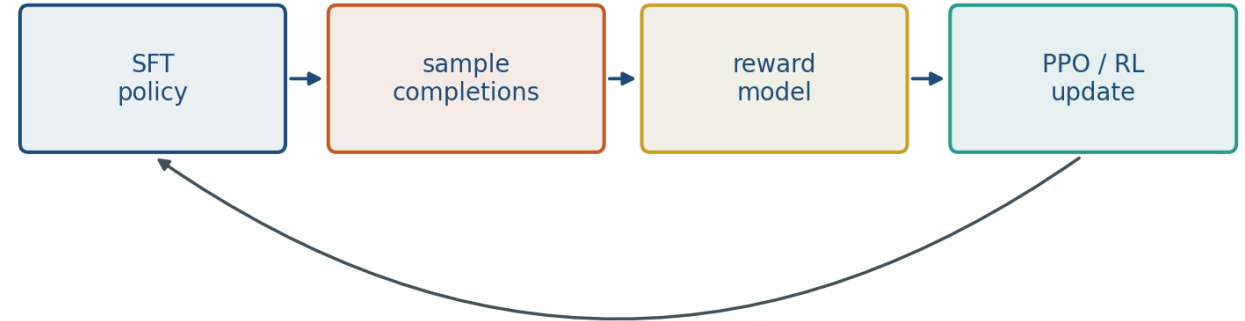
- What / why. After SFT and an RM, you sample from the current policy, score with r , update to raise reward minus KL to a frozen reference. SFT does not explore. Best-of- n uses r only at decode. RLHF trains π to look high-reward as it samples.
- Architecture. Policy π_θ , reference π_{ref} (usually SFT), RM r (frozen in the PPO stage), optional critic.
- Formula. $\max_{\pi} \mathbb{E}[r(x, y)] - \beta \text{KL}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))$.
- Tradeoffs. + Can beat SFT on preference evals. – Unstable, expensive, RM misspecification; people look to DPO (next note) to skip PPO.

How. PPO clip + advantage. $\beta = 0$ (no KL) is reward hacking: the policy exploits r .

SFT, then RM, then a policy update

- Stages 1–2 use a fixed dataset.
- Stage 3 needs fresh samples from the current π_θ .
- CS224N L10: InstructGPT scaled this stack.

RLHF: climb the reward, stay near the SFT policy (KL penalty)



Climb reward, stay near SFT

$$J = r_{\phi}(x, y) - \beta \text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}})$$

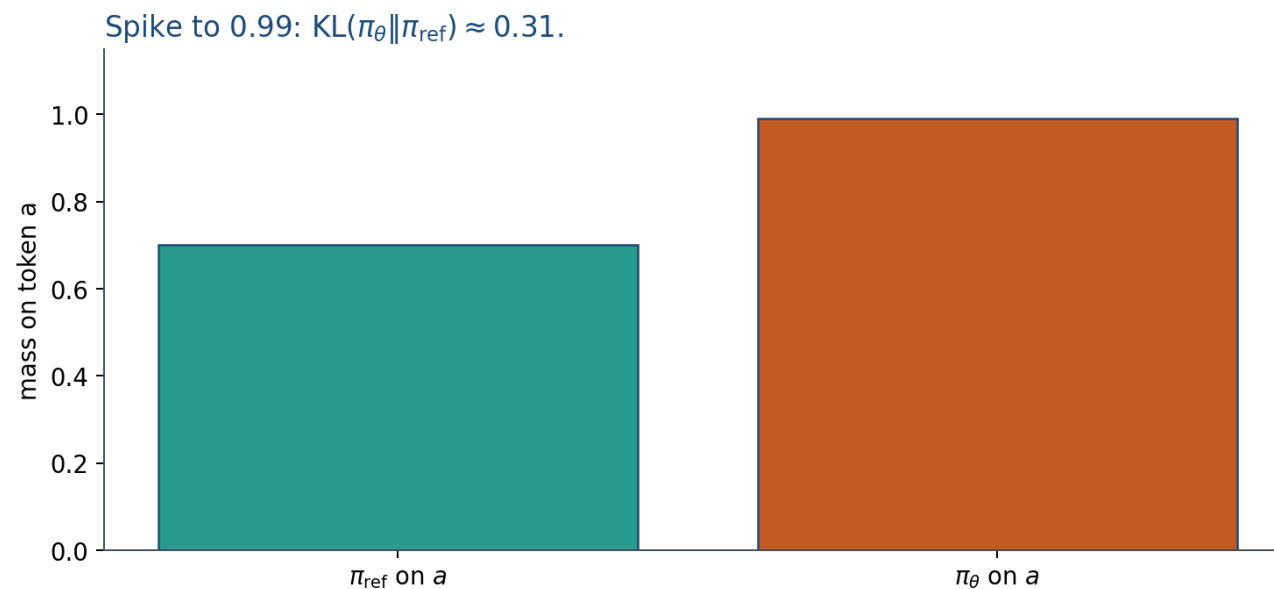
$\pi_{\text{ref}} = \pi_{\text{SFT}}$, frozen.

Large β stays near SFT. $\beta = 0$ is no leash.

You do not derive PPO here. You write the leash.

A two-word vocab

- $\pi_{\text{ref}} = (0.7, 0.3)$, $\pi_{\theta} = (0.99, 0.01)$.
- $\text{KL} \approx 0.31$.
- $\beta = 0.1$ subtracts 0.031. $\beta = 0$ subtracts nothing.



Who wins the objective

$\beta = 0$

Reward rises. KL explodes. Text becomes a caricature of the raters.

Large β

The policy barely leaves SFT. You paid for PPO and bought almost nothing.

What to take from this lecture

On-policy

Last week's samples are stale once π_θ moved.

Leash

π_{ref} is SFT, because the RM was trained on SFT-style answers.

Project

Best-of-n or DPO. Full PPO on an LLM is usually out of scope.